

Learner Assessment Brief

AI Concepts for Official Statisticians

OS-A02-A09 | Validated Candidate Version 1.0 | 26 September 2026

Assessment decision

You must demonstrate that you can classify an AI mechanism, decompose an official-statistics task, design credible evidence, diagnose failure propagation and defend a bounded institutional decision. Technology enthusiasm, fluent prose and aggregate accuracy do not substitute for this evidence.

Element	Requirement
Terminal product	AI Task and Technology Classification Dossier.
Written submission	Maximum 3,500 words plus evidence appendices.
Practical evidence	Executed or reproducible analysis using only approved data.
Oral defence	15 minutes plus assessor questions.
Competence	At least 70/100, no critical failure and at least half in each mandatory criterion.
Lead level	At least 85/100 with independent challenge, credible alternatives and defensible uncertainty.
Submission boundary	Do not include unapproved confidential or identifiable information.

Authenticity

Your submission must identify any AI assistance, prompts, model/version where known, retrieved sources, verification performed, corrections made and decisions retained by you. Fabricated execution or validation is a critical failure.

Terminal scenario

The Kintaba National Statistics Office is considering an AI-enabled workflow for one of five contexts: occupation coding, record linkage, crop-area estimation, release drafting or public statistical query support. Select one context, or propose an authorized equivalent. Your task is to determine what, if anything, should be permitted now.

Required dossier sections

Section	Evidence
1 Purpose and boundary	Product, user, population, consequence, exclusions and authority.
2 Current baseline	Existing workflow, quality, time, burden and failure modes.
3 Task decomposition	Operations, learned inference, judgement and release authority.
4 Mechanism classification	Observable mechanism, adaptation and uncertainty.
5 Evaluation design	Independent reference, sampling, metric, denominator, tolerance and subgroups.
6 Results or test plan	Executed results or clearly labelled prospective protocol.
7 Failure propagation	Effects on statistics, users, groups and institutional duties.
8 Controls	Prevention, detection, containment, escalation, stopping and rollback.
9 Decision	Permit bounded pilot, redesign, reject, pause or escalate.
10 Residual uncertainty	Unknowns, owners, monitoring and review date.

Required baseline and AI assisted comparison

Perform the same bounded task with the approved baseline and with the candidate AI-assisted workflow. If execution is not authorized, produce a prospective evaluation protocol and do not imply that testing occurred.

Comparison field	Baseline	AI assisted
Inputs and population		
Procedure and version		
Elapsed time		
Verification time		
Corrections and rework		
Critical failures		
Usable output		
Residual uncertainty		

Quality adjusted efficiency

Report total elapsed time together with verification, correction, escalation and unusable-output costs. A faster first draft is not an efficiency gain when end-to-end quality work increases or a critical error remains.

Evidence and verification log

ID	Claim or output	Source or reference	Verification	Correction	Disposition
E1					
E2					
E3					
E4					
E5					

Prompt and configuration record

Field	Record
Tool or model and version	
Date and configuration	
Prompt or rule set	
Retrieved sources or data	
Variability test	
Human decisions retained	

Attach longer prompts, code, outputs and source extracts as versioned appendices. Never paste protected data into an unapproved service.

Failure and control analysis

Failure event	Immediate effect	Downstream consequence	Affected group	Control	Stopping owner

Mandatory challenge questions

- Which evidence could invalidate your preferred decision?
- Which subgroup or rare case could be hidden by the aggregate result?
- What alternative method is credible?
- Who has authority to approve, stop and release?
- What would trigger reassessment?

Assessment rubric disclosed to learners

Criterion	Weight	Evidence expected
Purpose and boundary	10	Specific official-statistics use and exclusions.
Task decomposition	15	Operations, judgement and authority separated.
Mechanism classification	15	Correct mechanism with uncertainty.
Evidence and assumptions	15	Independent reference, baseline, population, metric and tolerance.
Failure propagation	20	Material chain to statistic, decision, group or duty.
Controls and authority	15	Prevention, detection, containment, stopping and ownership.
Decision and communication	10	Bounded status, conditions and residual uncertainty.

Critical failures

- Unapproved confidential or identifiable data.
- Fabricated validation, sources or executed evidence.
- Delegation of statistical release authority to the system.
- Approval based only on fluency, citations or aggregate accuracy.
- Omission of a known material invalidity or affected subgroup.
- No stopping or escalation route for a consequential system.

Oral defence

Stage	Minutes	Requirement
Opening decision	2	State scope, decision and strongest evidence.
Evidence defence	5	Explain baseline, reference, tolerances and results.
Challenge	5	Respond to counterexample, subgroup and failure questions.
Revision	3	State what would change the decision and how.

Permitted materials

- Submitted dossier and appendices.
- Your executed notebook or reproducible evidence.
- Controlled source documents supplied for the assessment.

Not permitted

- New unrecorded AI-generated answers during the defence.
- Unapproved external data or protected information.
- Claims of tests that cannot be reproduced or inspected.

Submission checklist

Check	Confirmed
Dossier uses the controlled template and identifier.	
All AI assistance and prompts are disclosed.	
Baseline comparison and quality-adjusted time are reported.	
Every material claim has a verification record.	
Subgroups, rare cases and uncertainty are addressed.	
Decision authority and stopping rule are named.	
No unapproved protected information is included.	
Files open, code runs and references resolve.	

Reassessment

When evidence is incomplete but no critical failure occurred, reassessment normally repeats only the failed evidence. Systemic reasoning failure or any critical failure may require full reassessment under the authorized procedure.

Controlled source basis

- OS-A02-A01 Canonical Lesson v0.2.
- OS-A02-A06 Executable Classification Laboratory v1.0.
- OS-A02-A07 Practice Exercises v1.0.
- OS-A02-A08 Solutions and Decision Guide v1.0.